

AC2204 Marking Scheme

Introduction to Management Accounting · Autumn 2023 · Paper 1

Examination Session and Year	Autumn 2023
Module Code	AC2204
Module Title	Introduction to Management Accounting
Paper Number	Paper 1
External Examiner	Professor Martin Quinn
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided they are not programmable and cannot store text (other than algebraic notation).

Instructions to Candidates (as per exam paper): Answer three questions in total. Attempt Question One from Section A (compulsory) and TWO questions from Section B.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Q1 (compulsory)	Split Ltd	Marginal vs absorption costing, profit reconciliation, management vs financial accounting	40
Q2 (Section B option)	Morgan Ltd	Traditional (absorption) costing vs activity-based costing, cost behaviour theory	30
Q3 (Section B option)	Crisper Ltd	High-low method, cost estimation techniques, management accounting terminology	30
Q4 (Section B option)	Statement evaluation	Breakeven, margin of safety, operating leverage, contribution margin ratio, multi-product CVP	30

QUESTION 1**Split Ltd**

Total: 40 marks

Split Ltd commenced operations on 1 January 2021 and manufactures and sells a single product. Candidates are required to (A) prepare income statements under both marginal and absorption costing, (B) explain and reconcile the resulting profit difference, and (C) identify three differences between management accounting and financial accounting.

Part	Requirement	Marks
(A)	Unit product cost, closing stock value and operating profit – marginal and absorption costing	18
(B)	Explanatory note and profit reconciliation	10
(C)	Three differences between management accounting and financial accounting	12

Question 1(A)**18 marks**

Split Ltd commenced operations on 1 January 2021. Sales 5,240 units; Production 6,800 units; Selling price per unit €34; Variable manufacturing costs (total) €106,080; Fixed manufacturing costs (total) €59,160; Variable marketing costs (total) €19,340; Fixed marketing costs (total) €13,600. (A) Calculate the unit product cost, the value of closing stock and the operating profit using (i) Marginal costing and (ii) Absorption costing. Show the income statement for each method.

Full Model Answer**Preliminary workings**

Closing stock = Production – Sales = 6,800 – 5,240 = 1,560 units. Variable manufacturing cost per unit = €106,080 ÷ 6,800 = €15.60. Fixed manufacturing overhead per unit (absorption costing only, absorbed on units produced) = €59,160 ÷ 6,800 = €8.70.

(i) Marginal costing**Unit product cost and closing stock - marginal costing**

Item	€
Unit product cost (variable manufacturing cost only)	15.60
Closing stock value (1,560 units × €15.60)	24,336

Income statement - marginal costing

Item	€
Sales revenue (5,240 units × €34)	178,160
Less: Variable cost of goods sold (5,240 units × €15.60)	(81,744)
Less: Variable marketing costs	(19,340)
Contribution margin	77,076
Less: Fixed manufacturing costs	(59,160)
Less: Fixed marketing costs	(13,600)
Operating profit (marginal costing)	4,316

(ii) Absorption costing**Unit product cost and closing stock - absorption costing**

Item	€
Unit product cost (€15.60 variable + €8.70 fixed)	24.30
Closing stock value (1,560 units × €24.30)	37,908

Income statement - absorption costing

Item	€
Sales revenue (5,240 units × €34)	178,160
Less: Cost of goods sold (5,240 units × €24.30)	(127,332)
Gross profit	50,828
Less: Variable marketing costs	(19,340)
Less: Fixed marketing costs	(13,600)
Operating profit (absorption costing)	17,888

Mark Allocation

Marks	Criteria
16-18	Both unit product costs, both closing stock values, and both fully-formatted income statements correct, with contribution margin and gross profit subtotals clearly shown.
12-15	Substantially correct with a minor computational error in one figure, or one method fully correct and the other largely correct.
7-11	Correct approach to at least one costing method; some errors or omissions in workings or presentation.
0-6	Limited understanding of the distinction between the two costing methods; minimal correct workings.

Question 1(B)**10 marks**

The managing director is confused by the figures calculated in Part A above and does not understand how it is possible to have two profit figures and two values for units in closing stock. Prepare a short note that explains why this is happening. Include in your answer calculations (profit reconciliation) which explain the difference in the operating profit figures resulting from the use of the two costing methods in Part A above.

Full Model Answer**Why the two methods give different closing stock values**

Marginal costing treats fixed manufacturing overhead as a period cost, charged in full against the period in which it is incurred, so only the variable manufacturing cost (€15.60) is included in the value of each unit in stock. Absorption costing treats fixed manufacturing overhead as a product cost, which is absorbed into each unit produced (€8.70 per unit) and carried forward in stock until the unit is sold. Because 1,560 units of production remain unsold at the year end, absorption costing carries an additional €8.70 per unit (€13,572 in total) of fixed overhead forward in the closing stock value, whereas marginal costing does not.

Why the two methods give different profit figures

Because production (6,800 units) exceeded sales (5,240 units) in the period, more fixed manufacturing overhead was carried forward into closing stock under absorption costing than was released from opening stock (there was no opening stock, as this was the company's first year of operations). This means less fixed overhead was charged against this period's profit under absorption costing than under marginal costing, resulting in a **higher reported profit under absorption costing** whenever production exceeds sales.

Profit reconciliation

Item	€
Operating profit – marginal costing	4,316
Add: Fixed manufacturing overhead carried forward in closing stock (1,560 units × €8.70)	13,572
Less: Fixed manufacturing overhead carried forward in opening stock (none – first year of trading)	0
Operating profit – absorption costing	17,888

General rule: When production exceeds sales, absorption costing reports a higher profit than marginal costing (as here). When sales exceed production, marginal costing reports a higher profit than absorption costing (more fixed overhead is released from opening stock than is deferred in closing stock). When production equals sales, the two methods report identical profit.

Mark Allocation

Marks	Criteria
9-10	Clear explanation of the fixed overhead deferral in stock under absorption costing, why production exceeding sales causes profit to differ, and a correct, clearly presented profit reconciliation.
6-8	Good explanation with a correct or substantially correct reconciliation; some lack of clarity or a minor computational error.
3-5	Some understanding shown of why the figures differ, but explanation is generic or the reconciliation is materially incomplete/incorrect.
0-2	No meaningful explanation or reconciliation provided.

Question 1(C)**12 marks**

Identify and describe three differences between management accounting and financial accounting.

Full Model Answer**1. Users of the information**

Financial accounting produces reports primarily for external users — shareholders, lenders, tax authorities, and other outside parties — who need a reliable overall picture of the entity's performance and position. Management accounting produces reports primarily for internal users — managers at all levels within the organisation — who need detailed information to plan, direct, and control day-to-day operations and make decisions.

2. Regulation and format

Financial accounting is governed by external regulatory requirements (company law, accounting standards such as IFRS/GAAP), which prescribe the format, content and disclosure requirements of the financial statements to ensure comparability across entities. Management accounting is not subject to any external regulatory framework; reports can be prepared in whatever format and level of detail management finds most useful, tailored to the specific decision at hand.

3. Time orientation

Financial accounting is primarily historical, reporting on transactions and performance that have already occurred, typically summarised annually or at other fixed reporting intervals. Management accounting is forward-looking as well as historical, placing significant emphasis on budgeting, forecasting and planning for future periods to support decision-making, in addition to reporting on past performance.

Other acceptable differences: Candidates may alternatively discuss reporting frequency (financial accounting: periodic, typically annual/interim; management accounting: as frequently as needed, e.g. daily/weekly), or scope/level of detail (financial accounting: whole-entity aggregated figures; management accounting: detailed segment, product, or departmental information). Any three well-explained, distinct differences receive full credit.

Mark Allocation

Marks	Criteria
10-12	Three clearly identified and well-explained differences, each distinct and correctly characterising both management and financial accounting.
7-9	Three differences identified with reasonable explanation, or two differences explained in excellent depth.
4-6	Two differences identified with limited explanation, or several generic points made without clear distinction.
0-3	Minimal or no relevant differences identified.

QUESTION 1 TOTAL — 40 MARKS

QUESTION 2**Morgan Ltd**

Total: 30 marks

Morgan Ltd currently uses traditional (absorption) costing, absorbing production overheads of €252,000 on the basis of machine hours across three products. Candidates are required to (A) recalculate product costs under both traditional and activity-based costing, and (B) distinguish between variable, fixed and mixed costs with the aid of diagrams.

Part	Requirement	Marks
(A)	Total costs and cost per unit – traditional costing and activity-based costing	21
(B)	Distinguish between variable, fixed and mixed costs, with diagrams and examples	9

Question 2(A)**21 marks**

Morgan Ltd absorbs production overheads (€252,000) on the basis of machine hours, at €36 per machine hour. Direct labour costs €27 per labour hour. Product A: 1,000 units, 2 DLH/unit, 2.0 MH/unit, 2,000 total MH, €25 DM/unit. Product B: 2,000 units, 3 DLH/unit, 1.5 MH/unit, 3,000 total MH, €40 DM/unit. Product C: 500 units, 4 DLH/unit, 4.0 MH/unit, 2,000 total MH, €10 DM/unit. ABC cost pools: Machining 30% (machine hours), Production scheduling 25% (set-ups), Quality control 30% (units), Materials handling 15% (purchase orders). Purchase orders: A=10, B=5, C=10 (total 25). Set-ups: A=5, B=20, C=10 (total 35). (A) Calculate the total costs and the cost per unit for each product, using (i) Conventional (absorption) costing and (ii) Activity-based costing.

Full Model Answer - (i) Traditional (absorption) costing

Overhead absorption rate = €252,000 ÷ 7,000 total machine hours (2,000 + 3,000 + 2,000) = €36 per machine hour, as given.

Traditional (absorption) costing - cost per unit

Item	Product A	Product B	Product C
Direct materials	25.00	40.00	10.00
Direct labour (DLH × €27)	54.00	81.00	108.00
Overhead (MH/unit × €36)	72.00	54.00	144.00
Total cost per unit	151.00	175.00	262.00

Traditional (absorption) costing - total costs

Item	Product A (€)	Product B (€)	Product C (€)
Units produced	1,000	2,000	500
Total cost (unit cost × units)	151,000	350,000	131,000

Check: Total overhead absorbed = €72,000 (A) + €108,000 (B) + €72,000 (C) = €252,000, agreeing with the total production overhead given.

Full Model Answer - (ii) Activity-based costing**Working - cost pools and activity rates**

Cost pool	Pool amount (€)	Cost driver	Total activity	Rate
Machining (30%)	75,600	Machine hours	7,000 MH	€10.80 per MH
Production scheduling (25%)	63,000	No. of set-ups	35 set-ups	€1,800.00 per set-up
Quality control (30%)	75,600	No. of units	3,500 units	€21.60 per unit
Materials handling (15%)	37,800	No. of purchase orders	25 orders	€1,512.00 per order
Total	252,000			

Working - overhead allocated to each product

Cost pool	Product A (€)	Product B (€)	Product C (€)
Machining (MH × €10.80)	21,600	32,400	21,600
Scheduling (set-ups × €1,800)	9,000	36,000	18,000
Quality control (units × €21.60)	21,600	43,200	10,800
Materials handling (orders × €1,512)	15,120	7,560	15,120
Total overhead	67,320	119,160	65,520

Check: Total overhead allocated = €67,320 + €119,160 + €65,520 = €252,000, agreeing with the total production overhead given.

Activity-based costing - total costs and cost per unit

Item	Product A	Product B	Product C
Direct materials (total)	25,000	80,000	5,000
Direct labour (total)	54,000	162,000	54,000
Overhead (from above)	67,320	119,160	65,520
Total cost	146,320	361,160	124,520
Units produced	1,000	2,000	500
Cost per unit	146.32	180.58	249.04

Comparison: ABC significantly increases the reported cost of Product A (€146.32 vs €151.00 under traditional costing is actually slightly lower here, while Product C falls from €262.00 to €249.04, and Product B rises from €175.00 to €180.58) — reflecting the fact that Product B and C consume disproportionately more set-up and materials-handling activity relative to their machine-hour usage than the traditional volume-based method recognises.

Mark Allocation

Marks	Criteria
18-21	Traditional costing correctly calculated; all four ABC cost pool rates correctly calculated; overhead correctly allocated to each product; ABC total cost and cost per unit correct for all three products.
13-17	Substantially correct with minor computational errors in one or two pool rates or allocations.
7-12	Traditional costing correct; ABC approach understood but rates or allocations contain multiple errors.
0-6	Limited understanding of either costing method; minimal correct workings.

Question 2(B)**9 marks**

Distinguish, with the aid of diagrams, between a variable cost, a fixed cost and a mixed cost. In your answer, give one example of each type of cost.

Full Model Answer**Variable cost**

A variable cost is a cost that varies, in total, in direct proportion to changes in the level of activity; the variable cost *per unit* remains constant. Graphed against volume, total variable cost is a straight line starting at the origin (zero cost at zero activity) and rising in proportion to volume. *Example: direct materials.*

Fixed cost

A fixed cost is a cost that remains constant, in total, regardless of changes in the level of activity within the relevant range; the fixed cost *per unit* decreases as volume increases. Graphed against volume, total fixed cost is a horizontal straight line. *Example: rental of factory premises.*

Mixed (semi-variable) cost

A mixed cost contains both a fixed element and a variable element; it does not start at zero and does not remain flat — it rises with volume but from a positive base cost incurred even at zero activity. Graphed against volume, total mixed cost is a straight line with a positive vertical intercept (the fixed element) and a positive slope (the variable element per unit). *Example: a landline telephone bill (fixed line rental plus variable call charges).*

Diagram guidance: Full marks require a simple sketch for each cost type: variable cost as a line through the origin; fixed cost as a horizontal line; mixed cost as a line with a positive y-intercept and positive slope, with the vertical axis labelled 'Total cost (€)' and the horizontal axis labelled 'Volume/Activity'.

Mark Allocation

Marks	Criteria
8-9	All three cost types correctly distinguished with accurate diagrams and one correct, distinct example each.
5-7	All three cost types correctly distinguished with correct examples; diagrams missing or only partially accurate.
2-4	One or two cost types correctly distinguished; diagrams or examples largely missing.
0-1	No meaningful distinction drawn between the three cost types.

QUESTION 2 TOTAL — 30 MARKS

QUESTION 3**Crisper Ltd**

Total: 30 marks

Crisper Ltd wishes to estimate its 2022 machine maintenance costs using six years of historical data on units manufactured and total machine maintenance costs. Candidates are required to (A) apply the high-low method, (B) list and explain two other methods of cost estimation, and (C) distinguish between three pairs of management accounting terms.

Part	Requirement	Marks
(A)	Estimate a cost function using the high-low method and forecast 2022 costs	10
(B)	List and explain two other cost estimation methods	8
(C)	Distinguish between three pairs of terms	12

Question 3(A)**10 marks**

Units manufactured and total machine maintenance costs, 2016–2021: 2016: 34,000 units, €290,100; 2017: 45,000 units, €344,000; 2018: 37,900 units, €309,210; 2019: 40,000 units, €319,500; 2020: 31,500 units, €277,850; 2021: 42,000 units, €329,300. (A) Using the High-Low method, estimate a cost function and use it to determine expected total machine maintenance costs for 2022, if 39,000 units are expected to be manufactured.

Full Model Answer

The high-low method identifies the highest and lowest *activity levels* (units manufactured) in the data set, not the highest and lowest costs.

Working - identify high and low points

Year	Units manufactured	Total cost (€)
2017 (highest activity)	45,000	344,000
2020 (lowest activity)	31,500	277,850
Difference	13,500	66,150

Working - variable cost per unit and fixed cost

Item	Value
Variable cost per unit ($€66,150 \div 13,500$ units)	€4.90 per unit
Fixed cost (at high point: $€344,000 - (45,000 \times €4.90)$)	€123,500
Fixed cost (at low point: $€277,850 - (31,500 \times €4.90)$) – check	€123,500

Cost function and 2022 estimate

Item	Value
Cost function	Total cost = $€123,500 + (€4.90 \times \text{units manufactured})$
Estimated 2022 cost at 39,000 units: $€123,500 + (€4.90 \times 39,000)$	$€123,500 + €191,100$
Estimated total machine maintenance cost, 2022	€314,600

Common error note: A frequent error is selecting the highest and lowest *cost* figures rather than the highest and lowest *activity* (units) figures — in this data set they happen to coincide (2017 is both the highest units and highest cost; 2020 is both the lowest units and lowest cost), so this particular data set does not penalise that error, but candidates should still be marked on having applied the correct method (selecting by activity level).

Mark Allocation

Marks	Criteria
9–10	High and low points correctly identified by activity level; variable cost per unit and fixed cost correctly calculated and cross-checked; cost function correctly stated and applied to give €314,600.
6–8	Correct method with a minor computational error, or the final estimate not cross-checked at both points.
3–5	High-low method attempted with the correct general approach but a significant error in identifying the high/low points or in the variable/fixed cost split.
0–2	Little or no understanding of the high-low method.

Question 3(B)**8 marks**

List and explain two other methods that can be used to estimate the fixed and variable components of a mixed cost.

Full Model Answer**1. Scattergraph (visual fit) method**

All historical data points (cost against activity level) are plotted on a graph, and a straight line is drawn through the data by visual judgement to best represent the relationship. The point where the line intersects the vertical axis gives an estimate of the fixed cost, and the slope of the line gives an estimate of the variable cost per unit. This method has the advantage of using all available data points, rather than only the two extreme points as in the high-low method, and allows outliers to be visually identified and excluded, but it relies on subjective judgement in fitting the line.

2. Regression analysis (least squares method)

Regression analysis uses a statistical technique to fit a line mathematically through all the data points, minimising the sum of the squared vertical distances between the line and each actual data point (hence "least squares"). This produces an objective, mathematically optimal estimate of both the fixed cost (the regression intercept) and the variable cost per unit (the regression slope coefficient), and additional statistics (such as the correlation coefficient, R^2) can be used to assess how well the cost function fits the underlying data. This is generally regarded as the most accurate and reliable of the cost estimation methods, but it requires more data and statistical software/technique to apply than the high-low or scattergraph methods.

Mark Allocation

Marks	Criteria
7-8	Two distinct methods correctly identified and well explained, including how each estimates the fixed and variable components.
5-6	Two methods identified with reasonable explanation, or one method explained in excellent depth.
2-4	One method identified with limited explanation.
0-1	No relevant methods identified.

Question 3(C)**12 marks**

Distinguish between each of the following terms widely used in management accounting: (i) Sunk costs and opportunity costs. (ii) Prime costs and conversion costs. (iii) Planning and control.

Full Model Answer**(i) Sunk costs and opportunity costs**

A sunk cost is a cost that has already been incurred in the past and cannot be recovered or changed by any current or future decision; because it does not differ between the alternatives being considered, it is irrelevant to decision-making. An opportunity cost is the potential benefit that is given up (foregone) when one alternative course of action is chosen over the next best alternative; it does not appear in the accounting records but is highly relevant to decision-making, since it represents a real economic sacrifice.

(ii) Prime costs and conversion costs

Prime cost is the total of all direct costs of production: direct materials plus direct labour (and any other direct expenses). Conversion cost is the cost of converting raw materials into a finished product: direct labour plus manufacturing (production) overhead. Direct labour is therefore common to both prime cost and conversion cost, but direct materials appears only in prime cost, and manufacturing overhead appears only in conversion cost.

(iii) Planning and control

Planning is the process of establishing objectives and determining the means by which those objectives will be achieved; it is forward-looking and includes activities such as budgeting and forecasting. Control is the process of evaluating actual performance against the plan (e.g. comparing actual results to budget), identifying variances, and taking corrective action where necessary; it is a monitoring and feedback process that follows on from, and feeds back into, the planning process.

Mark Allocation

Marks	Criteria
10-12	All three pairs of terms clearly and accurately distinguished with correct definitions of each term.
7-9	Two pairs distinguished clearly and accurately, with the third partially correct or less developed.
4-6	One or two pairs distinguished with reasonable accuracy; definitions are generic or incomplete.
0-3	Minimal or no accurate distinction drawn between the terms.

QUESTION 3 TOTAL — 30 MARKS

QUESTION 4**Statement Evaluation - CVP Analysis**

Total: 30 marks

Five independent statements relating to cost-volume-profit analysis, breakeven point, margin of safety, operating leverage and multi-product breakeven. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

Part	Topic	Marks
(A)	Oak Ltd – breakeven, margin of safety and operating leverage verification	6
(B)	Operating leverage and margin of safety in capital vs labour intensive companies	6
(C)	Fixed cost per unit behaviour within the relevant range	6
(D)	Contribution margin ratio vs margin of safety	6
(E)	Alex – multi-product breakeven with sales mix	6

Question 4(A)**6 marks**

Oak Ltd produces a single product. If total sales are 30,000 units, totalling €165,000; total variable expenses are €63,000 and total fixed costs are €56,100, then break-even point is 16,500 units, the margin of safety is 45% and operating leverage is 4.

Full Model Answer**PARTIALLY CORRECT**

The break-even point and margin of safety figures given in the statement are correct, but the operating leverage figure of 4 is incorrect — the correct figure is approximately 2.22. Overall, the statement should be assessed as **disagree**, since not all three figures asserted are correct.

Working

Item	Value
Selling price per unit ($€165,000 \div 30,000$)	€5.50
Variable cost per unit ($€63,000 \div 30,000$)	€2.10
Contribution margin per unit	€3.40
Break-even point ($€56,100 \div €3.40$)	16,500 units - correct as stated
Margin of safety ($((30,000 - 16,500) \div 30,000)$)	45% - correct as stated
Total contribution margin ($30,000 \times €3.40$)	€102,000
Operating profit ($€102,000 - €56,100$)	€45,900
Operating (degree of) leverage ($€102,000 \div €45,900$)	2.22 - NOT 4 as stated

Mark Allocation

Marks	Criteria
5-6	All three figures independently recalculated and verified; correct verdict of disagree, with the error in the operating leverage figure specifically identified and the correct value (2.22) given.
3-4	Break-even point and margin of safety correctly verified; operating leverage recalculated but the final verdict not clearly stated, or a minor computational error.
1-2	Some correct workings but the operating leverage error not identified.
0	No meaningful attempt to verify the figures.

Question 4(B)**6 marks**

Both the Operating Leverage and the Margin of Safety, at a given level of sales in a capital intensive company, will tend to be lower than in a labour intensive company, with the same total revenues and total expenses.

Full Model Answer**DISAGREE**

A capital intensive company typically has a higher proportion of fixed costs (and correspondingly lower variable costs) relative to a labour intensive company with the same total revenue and total expenses. A higher fixed cost base results in a **higher** operating leverage, not a lower one — the statement is incorrect on this point.

Operating leverage

Operating (degree of) leverage = Contribution margin ÷ Operating profit. A capital intensive cost structure (high fixed costs, low variable costs per unit) produces a higher contribution margin relative to operating profit at any given sales level than a labour intensive cost structure (low fixed costs, high variable costs), because more of each sales euro is retained as contribution before fixed costs are covered. This means operating leverage is **higher**, not lower, in the capital intensive company — profit is more sensitive to changes in sales volume.

Margin of safety

Margin of safety = (Actual sales – Break-even sales) ÷ Actual sales. A higher fixed cost base in a capital intensive company increases the break-even point (more contribution is needed to cover the larger fixed cost base), which reduces the margin of safety. On this point, the statement's direction (margin of safety tends to be lower in a capital intensive company) is **correct**.

Overall: The statement is only half right: margin of safety being lower in a capital intensive company is correct, but operating leverage being lower is incorrect (it should be higher). Because the statement asserts *both* are lower, and this is not true for operating leverage, the overall verdict is disagree.

Mark Allocation

Marks	Criteria
5-6	Correct identification that operating leverage is higher (not lower) in a capital intensive company, with correct reasoning linking fixed cost proportion to both operating leverage and margin of safety; correct overall verdict of disagree.
3-4	Correct reasoning on one of the two measures (operating leverage or margin of safety) with the other measure's direction incorrect or unexplained.
1-2	Some relevant understanding of cost structure and risk shown, but reasoning is generic or the verdict is unclear.
0	No meaningful engagement with the statement.

Question 4(C)**6 marks**

As production increases within the relevant range, fixed cost per unit decreases.

Full Model Answer**AGREE**

Total fixed costs remain constant in total within the relevant range of activity, regardless of the level of production. Because fixed cost per unit is calculated as total fixed cost divided by the number of units produced, as the number of units produced increases, the same total fixed cost is spread over more units, so the fixed cost per unit falls.

Illustration

For example, if total fixed costs are €100,000: at 10,000 units of production, fixed cost per unit = €10; at 20,000 units of production, fixed cost per unit = €5. The total fixed cost of €100,000 is unchanged in both cases, but the per-unit fixed cost has halved because it has been spread across twice as many units. This relationship only holds within the 'relevant range' — the range of activity over which the assumption of constant total fixed costs is valid; outside this range (e.g. requiring a new factory or additional supervisory staff), fixed costs may step up to a new, higher total.

Mark Allocation

Marks	Criteria
5-6	Correct agreement with clear explanation of why fixed cost per unit falls as volume rises (constant total fixed cost spread over more units), including an illustrative example and reference to the relevant range.
3-4	Correct agreement with reasonable explanation; illustrative example or relevant range qualification missing.
1-2	Correct agreement but with minimal or unclear explanation.
0	Incorrect verdict or no meaningful explanation.

Question 4(D)**6 marks**

The contribution margin ratio measures the amount by which sales revenue can fall before losses begin to be incurred.

Full Model Answer**DISAGREE**

The description given in the statement is the definition of the **margin of safety**, not the contribution margin ratio. The two are different (though related) CVP measures.

What the contribution margin ratio actually measures

The contribution margin ratio (contribution margin ÷ sales revenue, expressed as a percentage) measures the proportion of each euro of sales revenue that remains, after variable costs are covered, to contribute towards fixed costs and profit. It is used, for example, to quickly calculate the change in profit resulting from a given change in sales revenue, or to calculate break-even sales revenue directly (fixed costs ÷ contribution margin ratio).

What the statement is actually describing

The amount by which sales revenue can fall before the company begins to incur losses is the definition of the **margin of safety** (actual or budgeted sales less break-even sales, often expressed as a percentage of sales). The contribution margin ratio and the margin of safety are related — both are CVP measures — but they measure different things and are not interchangeable.

Mark Allocation

Marks	Criteria
5-6	Correct disagreement with a clear, accurate definition of the contribution margin ratio and a correct identification that the statement's description actually matches the margin of safety.
3-4	Correct disagreement with a reasonable definition of the contribution margin ratio; the margin of safety point not explicitly identified.
1-2	Correct disagreement but with an inaccurate or vague definition of the contribution margin ratio.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(E)**6 marks**

Alex makes two types of cakes. Budgeted cake sales: Cherry cake 18,000 units at €20.00, variable cost €12.00; Coconut cake 12,000 units at €15.00, variable cost €9.00. Based on the budgeted production and sales information, and assuming total fixed costs of €5,760, the breakeven point is 480 cherry cakes and 320 coconut cakes.

Full Model Answer**AGREE**

The multi-product breakeven figures given in the statement (480 cherry cakes, 320 coconut cakes) are correct.

Working - contribution margin per cake

Item	Cherry cake	Coconut cake
Selling price per unit	€20.00	€15.00
Variable cost per unit	€12.00	€9.00
Contribution margin per unit	€8.00	€6.00

Working - sales mix and weighted average contribution margin

Item	Value
Budgeted sales mix (18,000 : 12,000, simplified)	3 : 2 (cherry : coconut)
Contribution per 'basket' of 5 cakes ($3 \times €8 + 2 \times €6$)	€36.00
Weighted average contribution margin per unit ($€36.00 \div 5$)	€7.20

Working - breakeven point

Item	Value
Total breakeven units ($€5,760 \div €7.20$)	800 units
Cherry cake breakeven ($800 \times 3/5$)	480 units - correct as stated
Coconut cake breakeven ($800 \times 2/5$)	320 units - correct as stated

Mark Allocation

Marks	Criteria
5-6	Correct agreement with a full, correctly calculated working showing the weighted average contribution margin approach and both individual product breakeven figures verified.
3-4	Correct agreement with substantially correct workings; a minor computational error that does not change the overall verdict.
1-2	Correct verdict stated but with minimal or unclear supporting workings.
0	Incorrect verdict or no meaningful workings.

QUESTION 4 TOTAL — 30 MARKS